Advanced Topics in Natural Language Processing

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School of informatics

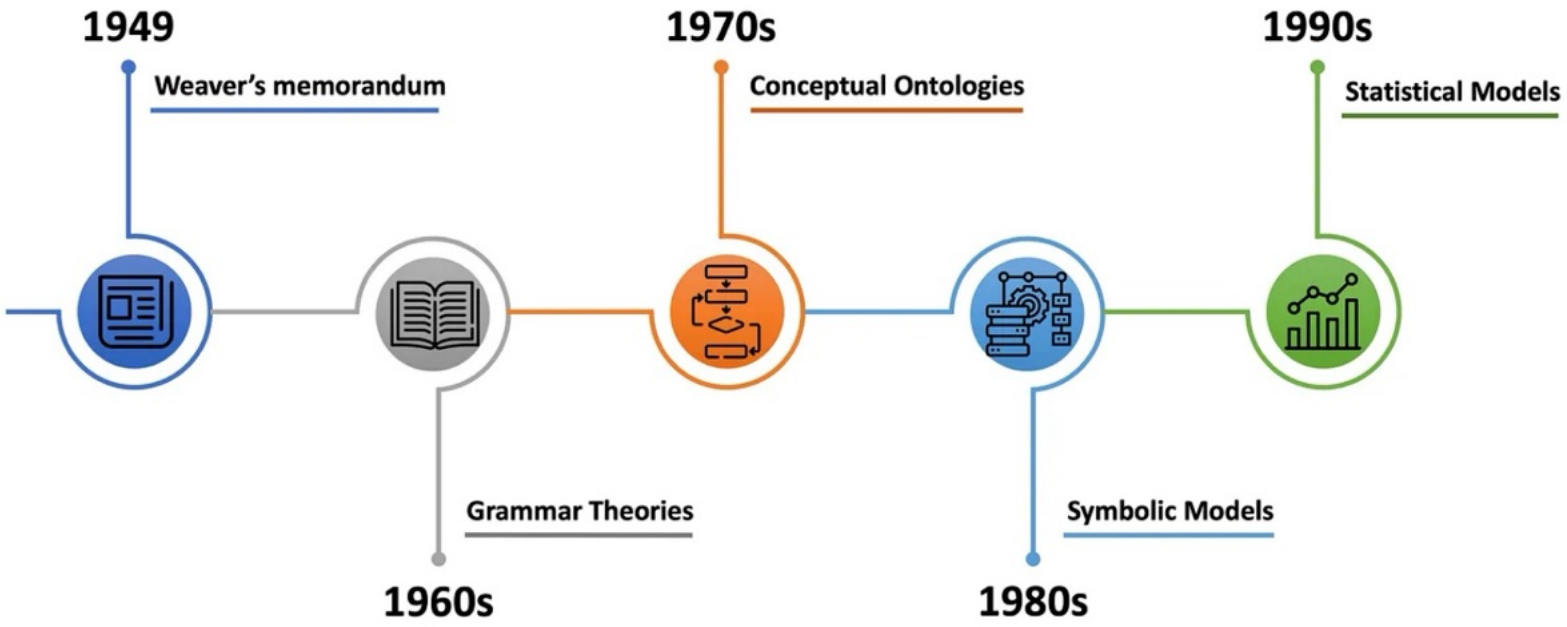
Outline

- Introduction
- Course content
- How the course will run

Introduction

AI and NLP

1963: Professor Donald Michie, established Europe's first AI research group in Edinburgh

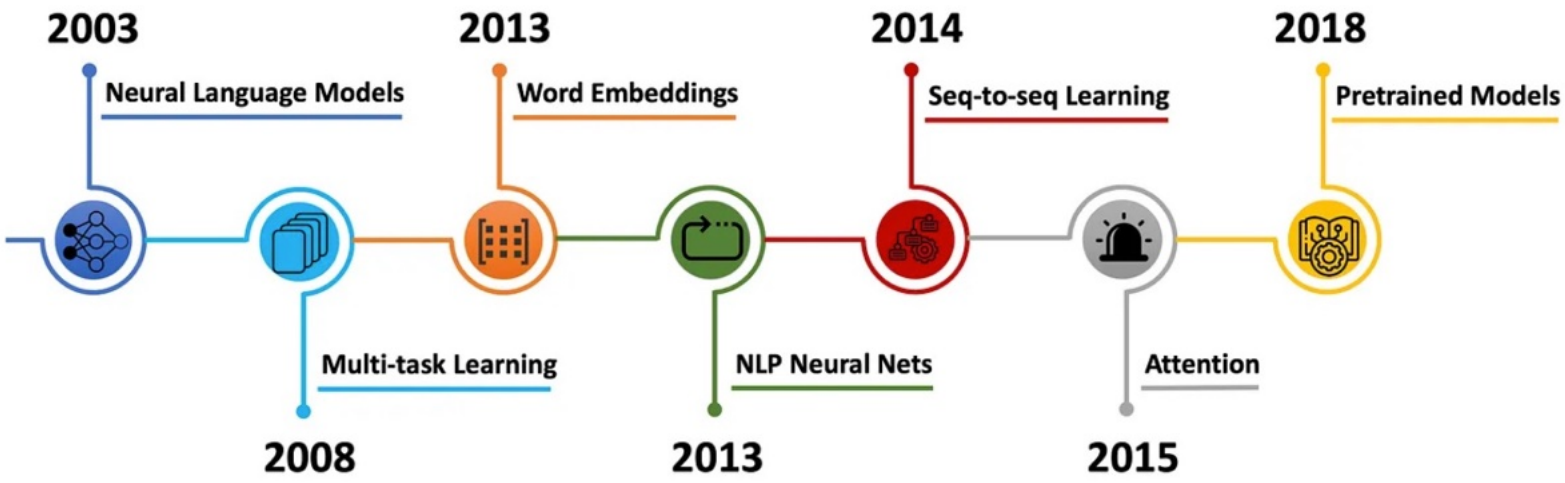


The big stages of NLP before the deep learning era.

Figure from https://medium.com/@antoine.louis/a-brief-history-of-natural-language-processing-part-1-ffbcb937ebce

AI and NLP

Neural Networks led to **paradigm shift** in NLP



The big stages of NLP in the deep learning era.

Figure from https://medium.com/@antoine.louis/a-brief-history-of-natural-language-processing-part-2-f5e575e8e37

Maturing transformer architectures, massive scaling and commercialization – NLP from niche to ubiquitous

Why now?

- **Architectures**: especially transformers capture long-range dependencies
- **Computational power** and large-scale **datasets**
- **Generative models**: self-supervised learning from raw text
- **Scaling** to larger models

Scaling

Doubling time:
Dataset size – 8 months
Training compute size - 5 months

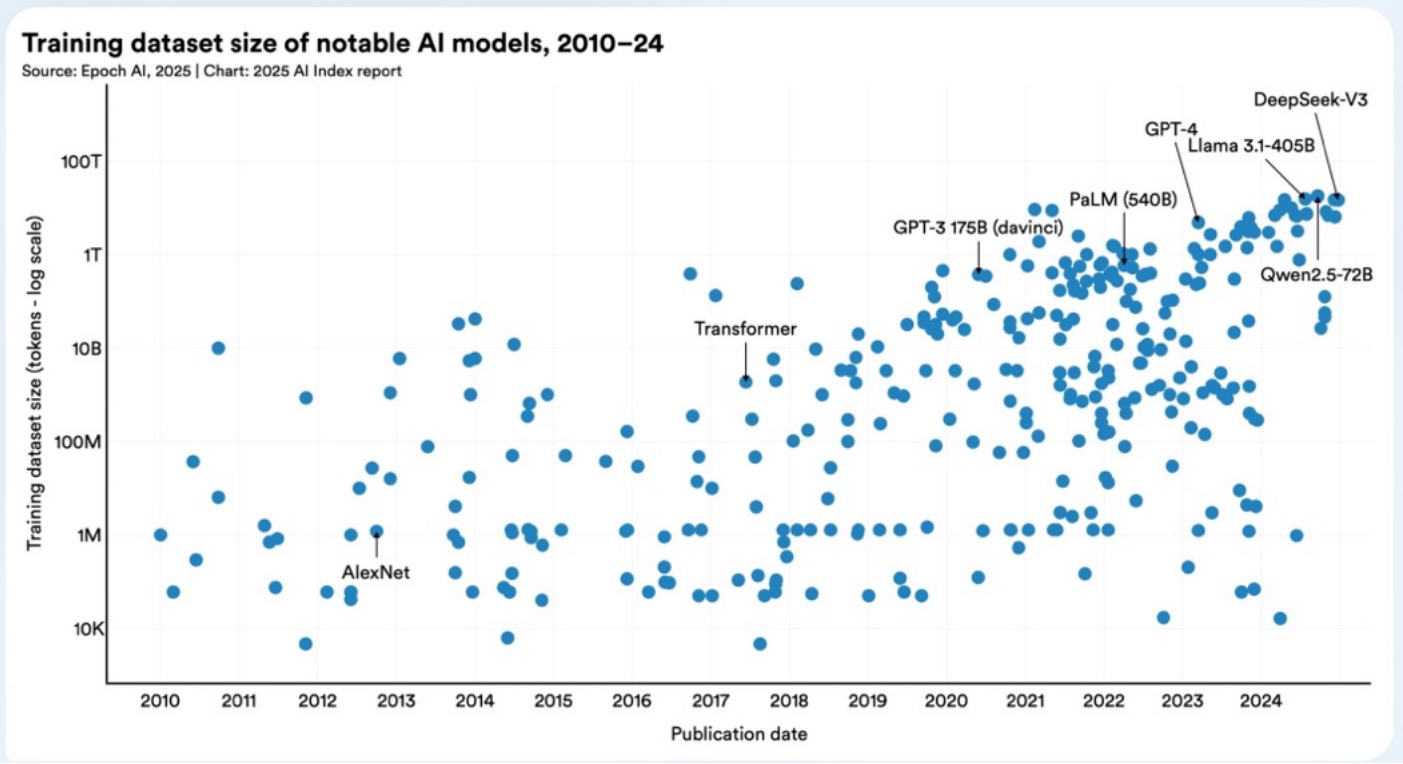
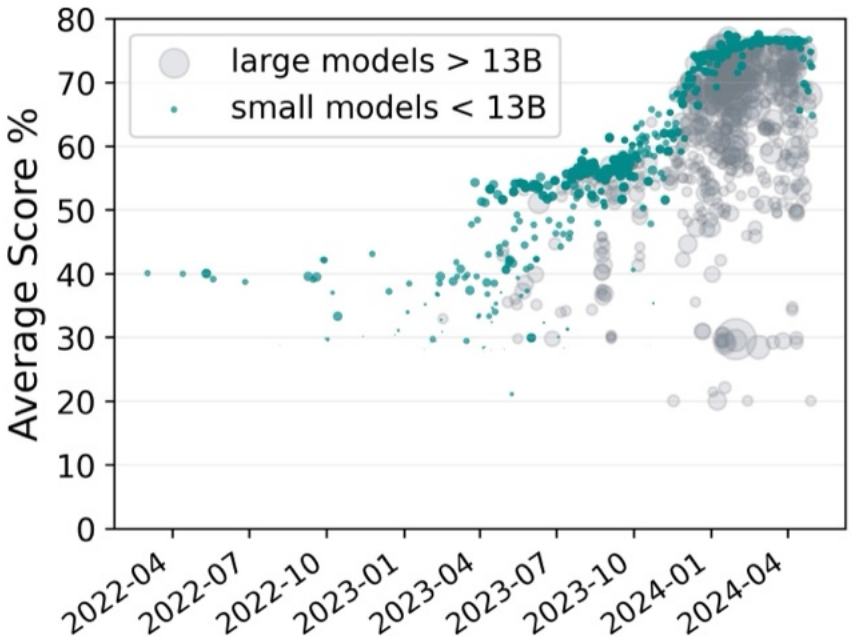
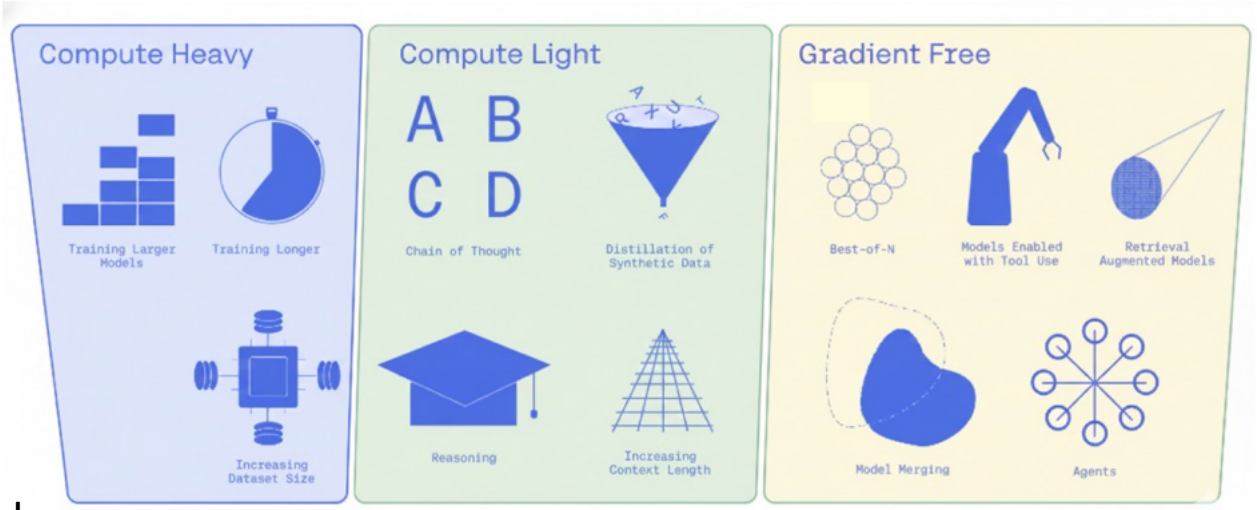


Figure from https://hai.stanford.edu/ai-index/2025-ai-index-report/research-and-development

Future of Scaling



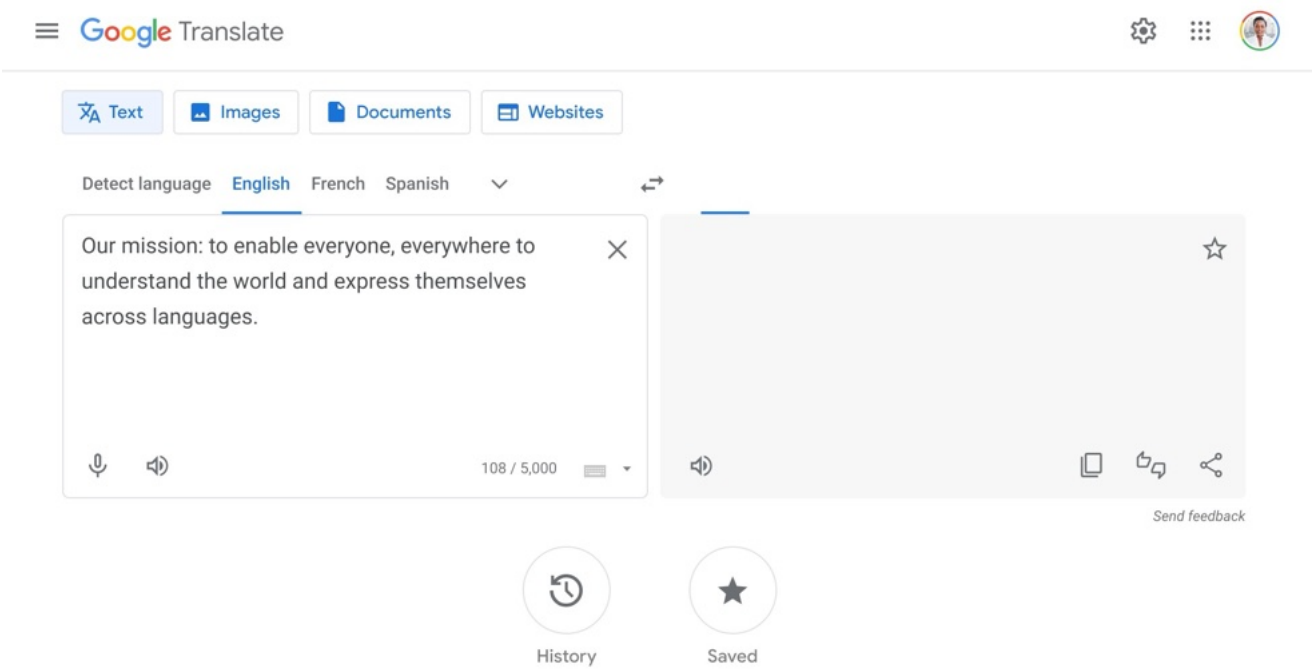


Many large models perform worse than smaller models
Eg. Qwen3-4B beats Gemma-27B on AIME'25

Sara Hooker (2025) [On the Slow Death of Scaling]

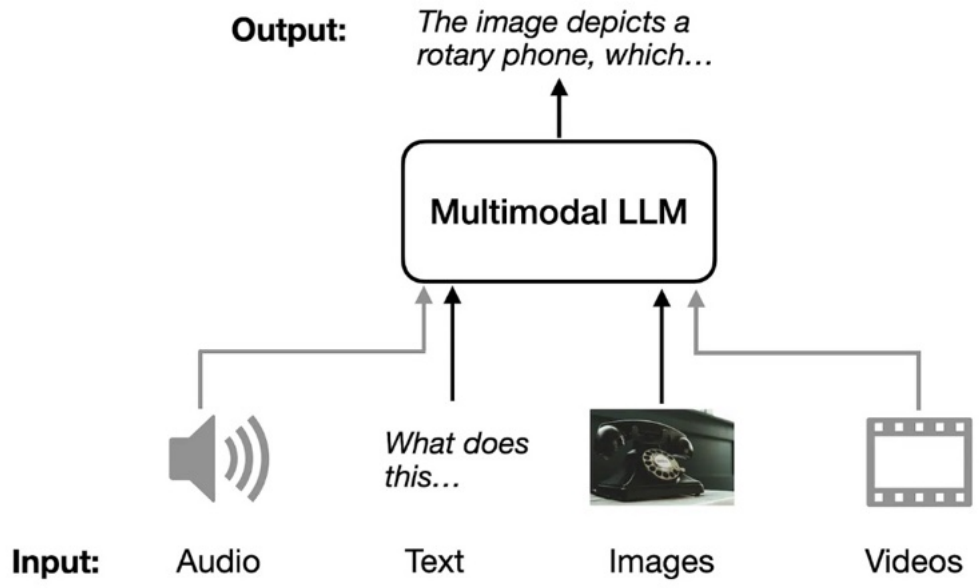
Cover far more languages and modalities

Google translate:
2004: English-Arabic
2026: 243 languages (including dialects and language varieties) with Palm2



https://translate.google.com/

Multimodal LLMs:



An illustration of a multimodal LLM that can accept different input modalities (audio, text, images, and videos) and returns text as the output modality.

https://magazine.sebastianraschka.com/p/understanding-multimodal-llms

Accelerating Improvement

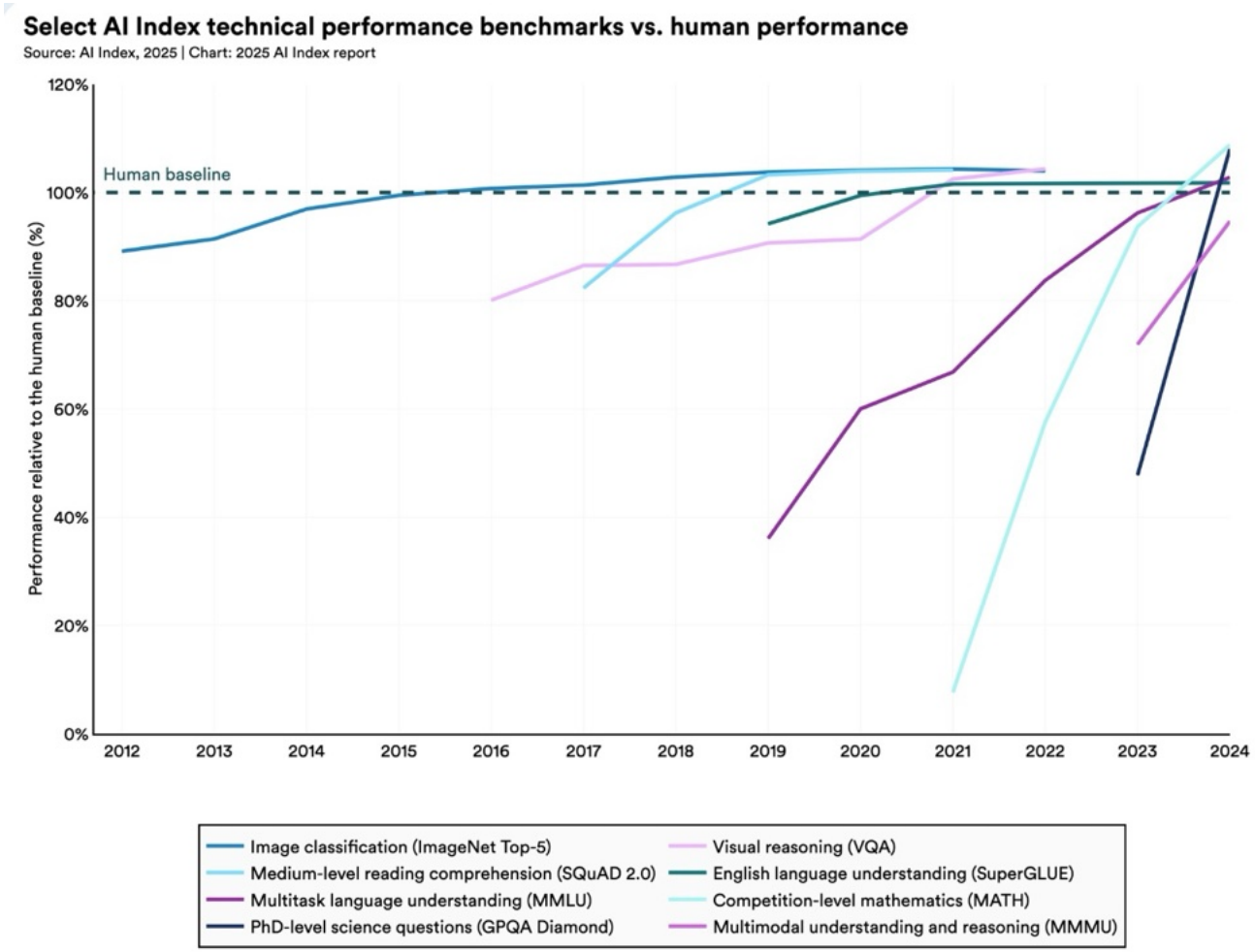


Figure from https://hai.stanford.edu/ai-index/2025-ai-index-report/technical-performance

Frontier tasks

GAIA

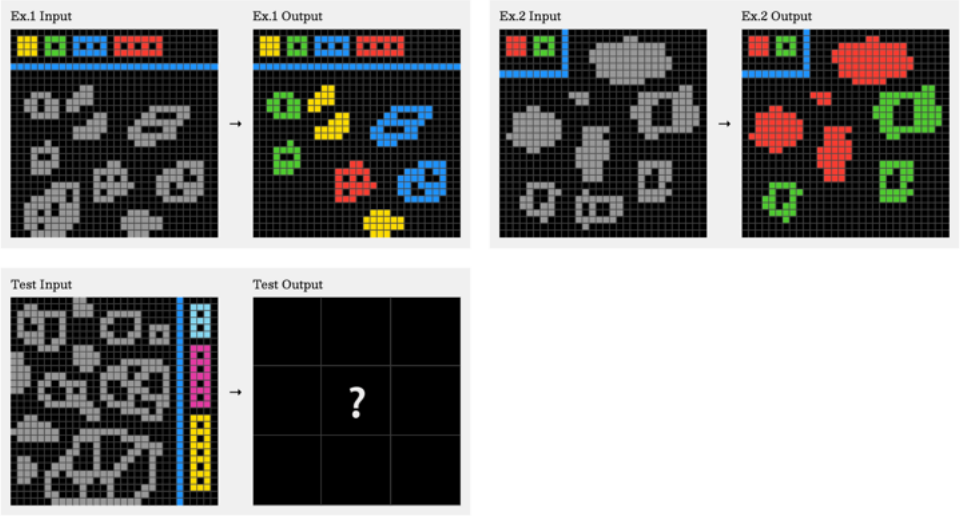
Level 3

Question: In NASA's Astronomy Picture of the Day on 2006 January 21, two astronauts are visible, with one appearing much smaller than the other. As of August 2023, out of the astronauts in the NASA Astronaut Group that the smaller astronaut was a member of, which one spent the least time in space, and how many minutes did he spend in space, rounded to the nearest minute? Exclude any astronauts who did not spend any time in space. Give the last name of the astronaut, separated from the number of minutes by a semicolon.
Ground truth: White; 5876

2023 GPT-4 15%, last month NVIDIA 87%

Mialon et al. (2023) Gaia: a benchmark for general ai assistants

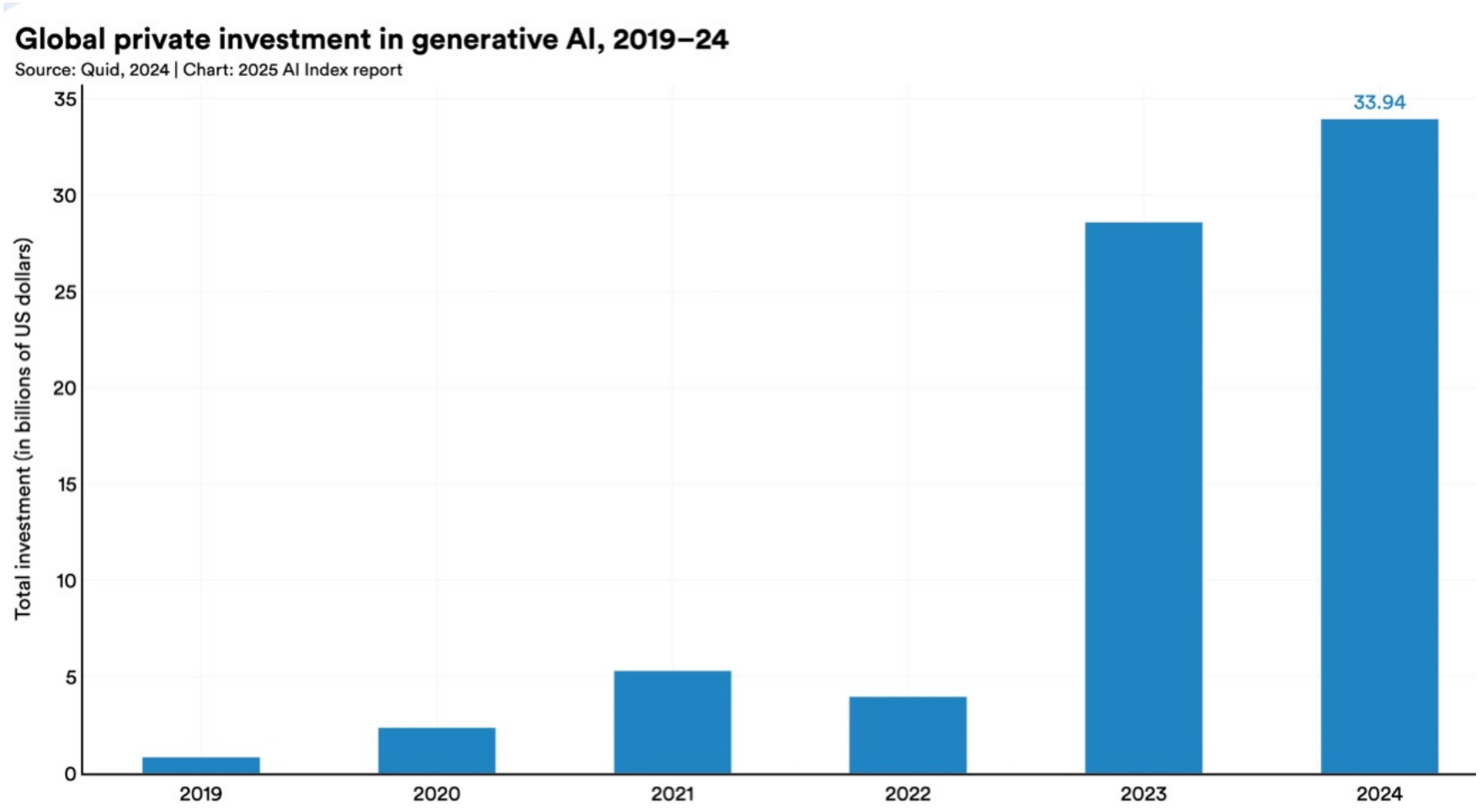
ARC-AGI-2



May 2025 O3mini 3%, last week Poetic 54%

Chollet et al. (2025) ARC-AGI-2: A New Challenge for Frontier AI Reasoning Systems

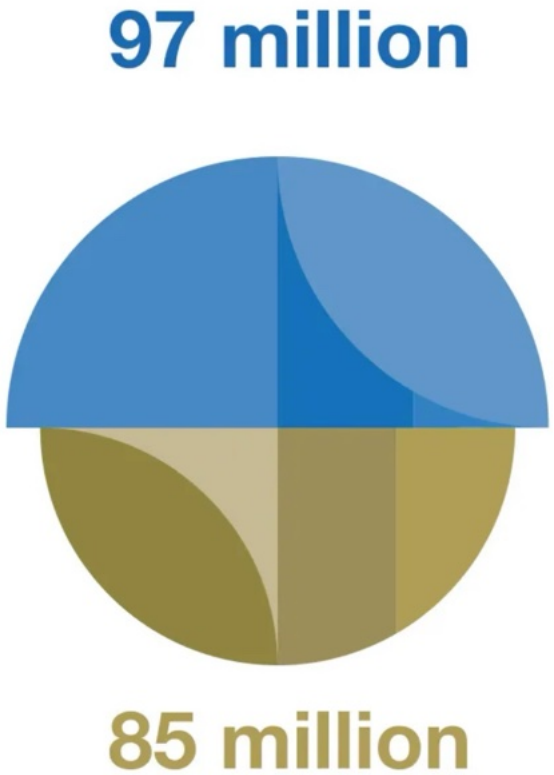
Economic Impact: growing investment



Bigger investment than: Manhattan, Apollo projects, dotcom boom

Economic Impact: jobs

New jobs created in 2025



Jobs lost in 2025

Growing job demand:

1. Data Analysts and Scientists
2. AI and Machine Learning Specialists
3. Big Data Specialists
4. Digital Marketing and Strategy Specialists
5. Process Automation Specialists
6. Business Development Professionals
7. Digital Transformation Specialists
8. Information Security Analysts
9. Software and Applications Developers
10. Internet of Things Specialists

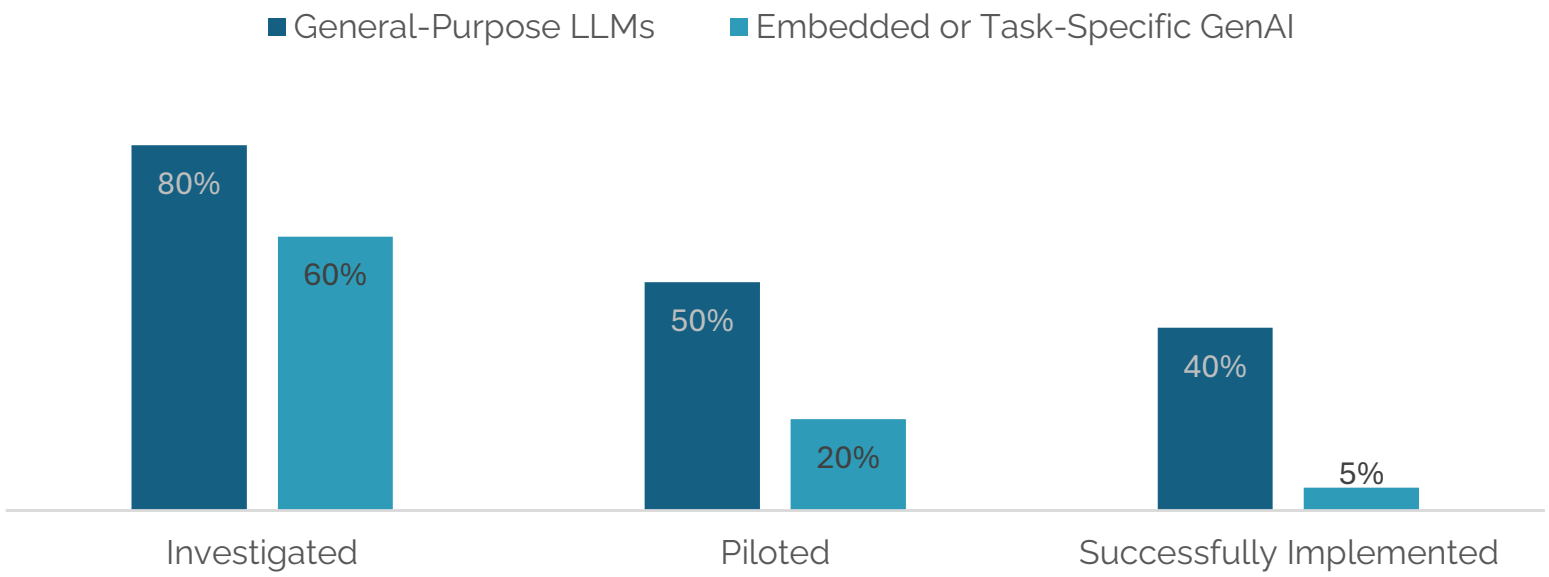
Decreasing job demand:

1. Data Entry Clerks
2. Administrative and Executive Secretaries
3. Accounting, Bookkeeping and Payroll Clerks
4. Accountants and Auditors
5. Assembly and Factory Workers
6. Business Services and Administration Managers
7. Client Information and Customer Service Workers
8. General and Operations Managers
9. Mechanics and Machinery Repairers
10. Material-Recording and Stock-Keeping Clerks

WEF emphasizes net positive effects through reskilling, with AI augmenting productivity

World Economic Forum https://sqmagazine.co.uk/ai-job-creation-statistics/

Economic Impact: real returns on investment still low

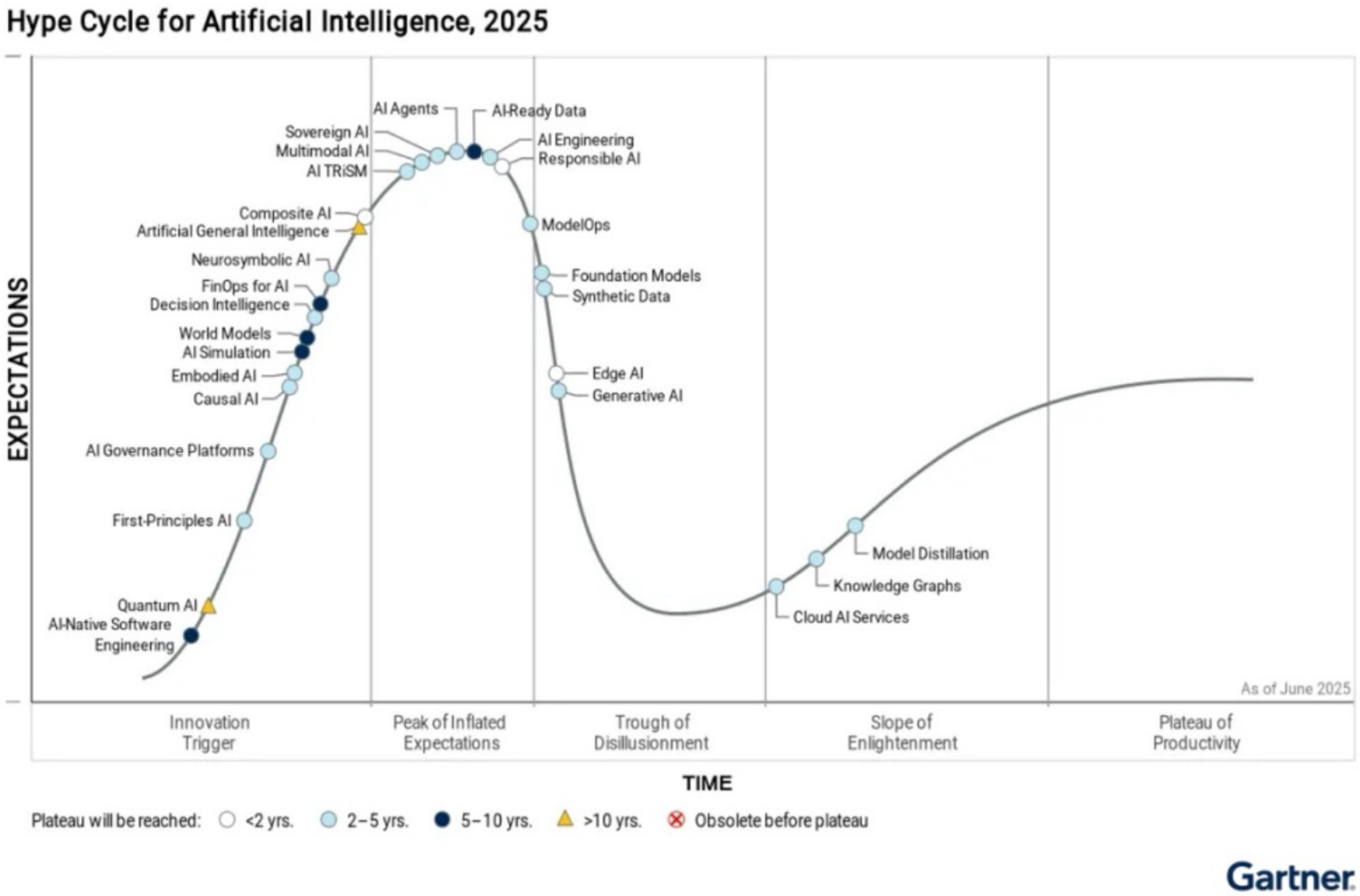


300 publicly disclosed AI Initiatives:

- 95% lean on generic tools, demos, brittle workflows – failed to deliver ROI
- 5% design for friction: high-value workflows, integrating deeply and shipping tools with memory and learning loops.
- Human friction – resistance, organization friction – policy, technical friction – lack learning

Challapally et al. [The GenAI Divide STATE OF AI IN BUSINESS 2025]

Economic Impact: Long Term value



https://buildingcreativemachines.substack.com/p/riding-the-gartner-hype-cycle-ai

Safety and Security Risks

- **Cyber-attacks**: tailored phishing methods or replicating malware.
- **Increased digital vulnerabilities**: new attack surfaces eg. prompt injection
- **Political and societal influence**: increase the scale, persuasiveness and frequency of disinformation
- **Insecure use and misuse**: integration into critical systems leading to biased and discriminatory systems
- **Erosion of trust in information**: pollution of the public information - hyper-realistic bots and deepfakes

https://www.gov.uk/government/publications/frontier-ai-capabilities-and-risks-discussion-paper/safety-and-security-risks-of-generative-artificial-intelligence-to-2025-annex-b

Meaning crisis: disinformation

AI images of Maduro capture reap millions of views on social media

Lack of verified information and advanced AI tools make it difficult to separate fact from fiction on US attack

Deep Fake Image – similar to reality – hard to verify – undermines confidence in what is real and what is not!

From https://www.theguardian.com/technology/2026/jan/05/maduro-venezuela-ai-images

Course content

Goals of ATNLP

- **Analyse** linguistic, machine learning, and ethical challenges in real-world NLP systems
- **Understand** and explain state-of-the-art NLP models and algorithms
- **Design**, **implement**, and adapt modern NLP systems for specific tasks
- **Evaluate** models rigorously, using error analysis to compare pros and cons of different approaches

Purpose of ATNLP

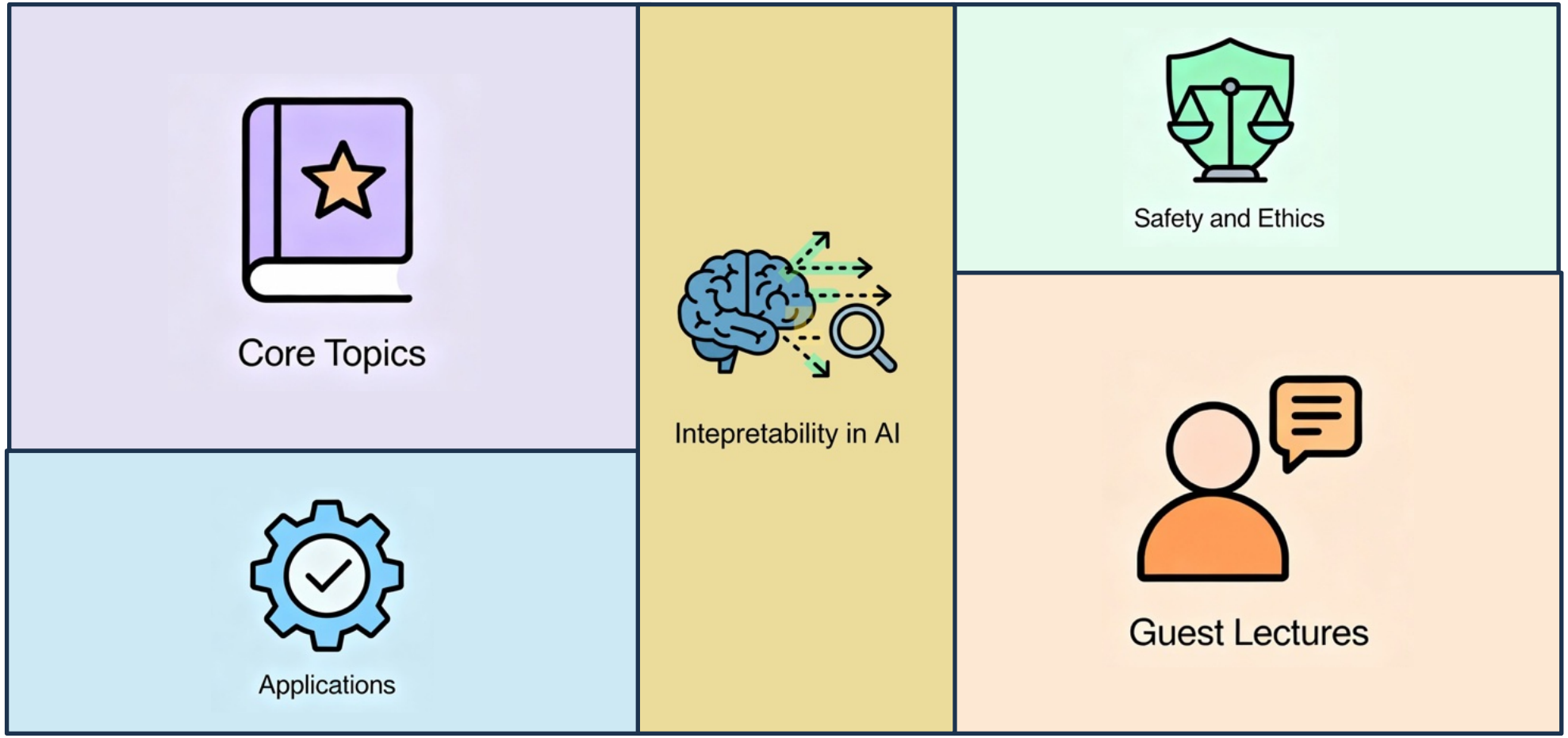
Technology's value lies solely in its power to transform our world in ways that enable better lives - Centre for Technomoral Futures

- Empowering responsible innovation: attention to fairness, transparency and real-world impact
- Focussing on evaluation: detect and mitigate harms
- Combining technical & critical reflection: **building future AI leaders**

Perspective on AI's role in helping human flourishing and avoiding the alternative:

Steven Bird (2025) Big AI is Accelerating the Metacrisis: What Can We Do?

Overview


Core Topics
Intepretability in AI
Safety and Ethics
Applications
Guest Lectures

Overview

Core Topics

Data, Evaluation, PEFT, Distillation, Long context, Reasoning, RAG, Multimodal, Multilingual

Intepretability in AI

Interpretability, formal machines, Parsing

Safety and Ethics

Safety and Security, Ethics

Guest Lectures

African languages, Sparsity, Diffusion, Explainability, Fact Checking, TBD

Applications

Translation, Summarisation, Question Answering, Agents

How the course will run

Required Preparation

Background required for the course:

- You should be familiar with Jurafsky and Martin (2025) Edition 3 from August 2025

- But this textbook serves as background only. Each lecture will rely on one or two papers as the main reading, or a chapter. Read them and discuss or ask questions. We will specify what sections to focus on.

- You will need solid maths: probability theory, linear algebra, some calculus. But this is not a maths course: we're not going to examine you on your ability to take derivatives. (LLMs can do that now anyway!)

Lecturers

Alexandra Birch

Shay Cohen

Alessandro Suglia

Teaching Assistants

Yifu Qiu

Helped with coursework design

Gustavo Cilleruelo

Helped with labs design

Both will be active on Piazza to help with technical questions and more

Course Infrastructure

- **OpenCourse:** Hosts all course materials (lecture slides, readings, tutorials, labs) and weekly schedule – signpost to everything else
- **Blackboard Learn**: Official Course information: links to everything (timetable/live lectures), assignments, contact information, announcements
- **Piazza**: Forum for posting questions about the course material; monitored by TAs and lecturers.
- **Gradescope**: Coursework submission system, linked with Learn. Used for coursework, includes plagiarism checking.
- **Dice**: Informatics computing environment, used for coursework. If you don't have a Dice account yet, apply for one through the ITO.

Course Mechanics

- **Friday** of every week: **readings** for the next week released; expectations for this week clarified. Lecture slides will often be available then too.
- **Three in-person lectures each week**; these are also live streamed and recorded for later viewing
- Most weeks there will one **guest lecture on Friday**
- You can ask questions in person during the lectures
- In each week, there is either a **tutorial** or a **lab session**

Guest Lectures

- Hear **from 8 leading researchers** shaping the future of NLP
- **Exposure** to cutting edge research areas beyond core curriculum
- Broadens perspective on **advanced, emerging topics and real-world innovation**
- These lectures are **not examinable** — designed for inspiration, curiosity, and exploration

Tutorial and Lab Sessions

- These run weeks 3 to 10 and **alternate weekly**. So each week has **either** a tutorial **or** a lab session.
- **Tutorials** are run in small groups led by a tutor. You should try to solve the tutorial exercises ahead of time, and then discuss your solutions in the tutorial session.
- **Lab sessions** are practical sessions in which you solve programming exercises (preparing for coursework and NLP projects).
- Labs are run in larger groups and are drop in. One lab demonstrators is on hand to help if you get stuck.
- Both tutorial and lab exercises are issued a week before – and **run in person**.

Tutorial and Lab Sessions

- If you're enrolled for this course, you will be automatically assigned a tutorial group and attendance is taken.
- If you are unhappy with your assigned groups, please use the Self Change functionality with the MyGroups section of Timetabler should be used to move groups
- If you have not yet been assigned a tutorial group, please contact the ITO.
- Labs start in week 3 and Tutorials start in week 4

Assessment: Exam

- Final exam worth 70% in April/May exam period (exact date not known yet)
- The final exam is timed and will consist of problem-solving questions.
- It will emphasise understanding and synthesis of ideas, rather than rote memorisation of technical details.

Assessment: Coursework

- A single coursework, worth 30%, issued 9 February, due 13 March.
- Coursework require you to implement and run code; the experiments can be time consuming, so start early!
- Coursework will include two main parts about using language models to reason: *prompt engineering* and *post-training (supervised fine-tuning and RL)*
- Prompt engineering will be done with OpenAI GPT models provided by the University ELM
- Post-training will require GPUs, but is possible to complete on the free-tier Google Colab
- Coursework are accompanied by office hours with the TA

More on Coursework

- Coursework can be done in pairs.
- This means you will work together with a classmate and submit a single solution.
- Both members of the pair will receive the same mark.
- You don’t have to work in a pair, but it’s strongly encouraged.
- Details on how pairs are formed will be released before the coursework (similar to other courses) – but you will be able to propose someone to pair up with
- Ideally, a pair will have people with slightly different backgrounds

How to get help

- Ask questions. Asking questions is how you learn.
- Answer questions. Answering questions helps you to learn!
  - **TA drop-in office hours**, starting week 2.
  - **Piazza forum**: course staff will answer questions once a day, Monday through Friday. You can answer questions any time! Your questions can be anonymous to classmates. They can be private (use this if the question pertains only to you).
  - **Don't ask us questions over email**. We might not see your question for days. And when we do, we will just repost it to Piazza, so that everyone can see the answer.

Summary

- Give students the knowledge and tools to immerse themselves in NLP
- A line up of guest lectures who make a diverse use of NLP and language models

- What's next? Applications: translation, summarisation and QA